Smart AgroBot for Soil Health Monitoring and Automated Farming using ESP32 and Firebase

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Table of Contents

Abstract *2*

1. Motivation and Background *2*

1.1 Context **2**

1.2 Motivation **2**

1.3 Problem Statement **2**

2. Literature Review and Introduction *3*

2.1 Related Work **3**

2.2 Research Gap **3**

2.3 Objective **3**

3. System Overview and Block Diagram *4*

3.1 System Architecture **4**

4. Hardware Implementation *5*

4.1 Components and Specifications **5**

5. Software Implementation *5*

5.1 Development Environment **5**

5.2 Firebase Database Structure **5**

5.3 Mobile Application **5**

6. Novelty and Innovation *6*

7. Sustainable Development Goals (SDG) Alignment *6*

8. Experimental Results and Observations *6*

9. Snapshots of Hardware *7*

10. Conclusion and Future Scope *8*

Key Achievements: **8**

Proposed Enhancements: **8**

11. References *9*

Abstract

The Smart AgroBot is an intelligent IoT-based mobile robotic platform engineered to automate agricultural monitoring and irrigation processes. Powered by the ESP32 microcontroller, it continuously measures environmental and soil parameters such as temperature, humidity, and moisture, transmitting this data to a Firebase Realtime Database. A custom-built mobile application designed with MIT App Inventor enables real-time visualization and control of robotic functions—including movement and irrigation—from remote locations.

The AgroBot operates on a four-wheel drive chassis controlled by an L298 motor driver and includes a relay-driven pump system for irrigation. The prototype enhances precision agriculture practices, optimizes water use, and significantly reduces manual labor. By prioritizing affordability and modular scalability, the Smart AgroBot provides an accessible entry point for small and medium-scale farmers into the realm of data-driven agricultural automation.

1. Motivation and Background

1.1 Context

Agriculture remains essential for global food security but continues to rely on manual and often outdated practices. Farmers face challenges such as inefficient irrigation, soil degradation, yield unpredictability, and water wastage. Traditional irrigation systems depend on schedule-based mechanisms, ignoring real-time soil and weather conditions, which results in water inefficiency and crop stress.

1.2 Motivation

The integration of automation and IoT technologies into agriculture presents an opportunity to revolutionize these traditional methods. However, most commercially available systems are costly or stationary, limiting their accessibility and functionality. The Smart AgroBot was conceptualized to address these gaps—creating a low-cost, mobile, and IoT-enabled robotic system capable of collecting soil data, uploading it to the cloud, and automating irrigation control wirelessly.

1.3 Problem Statement

1. To design and implement a mobile, low-cost, IoT-enabled system that can:

- Monitor essential soil and environmental parameters.
- Upload data to the Firebase cloud in real time.
- Permit remote manual control via a mobile application.
- Automate irrigation based on live sensor readings.

2. Literature Review and Introduction

2.1 Related Work

Existing studies and projects demonstrate varying degrees of smart farming automation:

- Arduino-based systems automate irrigation but lack mobility.
- Raspberry Pi solutions offer better computing but are expensive.
- IoT data loggers collect environmental data without actuation.

2.2 Research Gap

Previous efforts often lack seamless integration of sensing, actuation, and cloud interactivity. They rely on local storage or closed Wi-Fi networks, which limit field scalability.

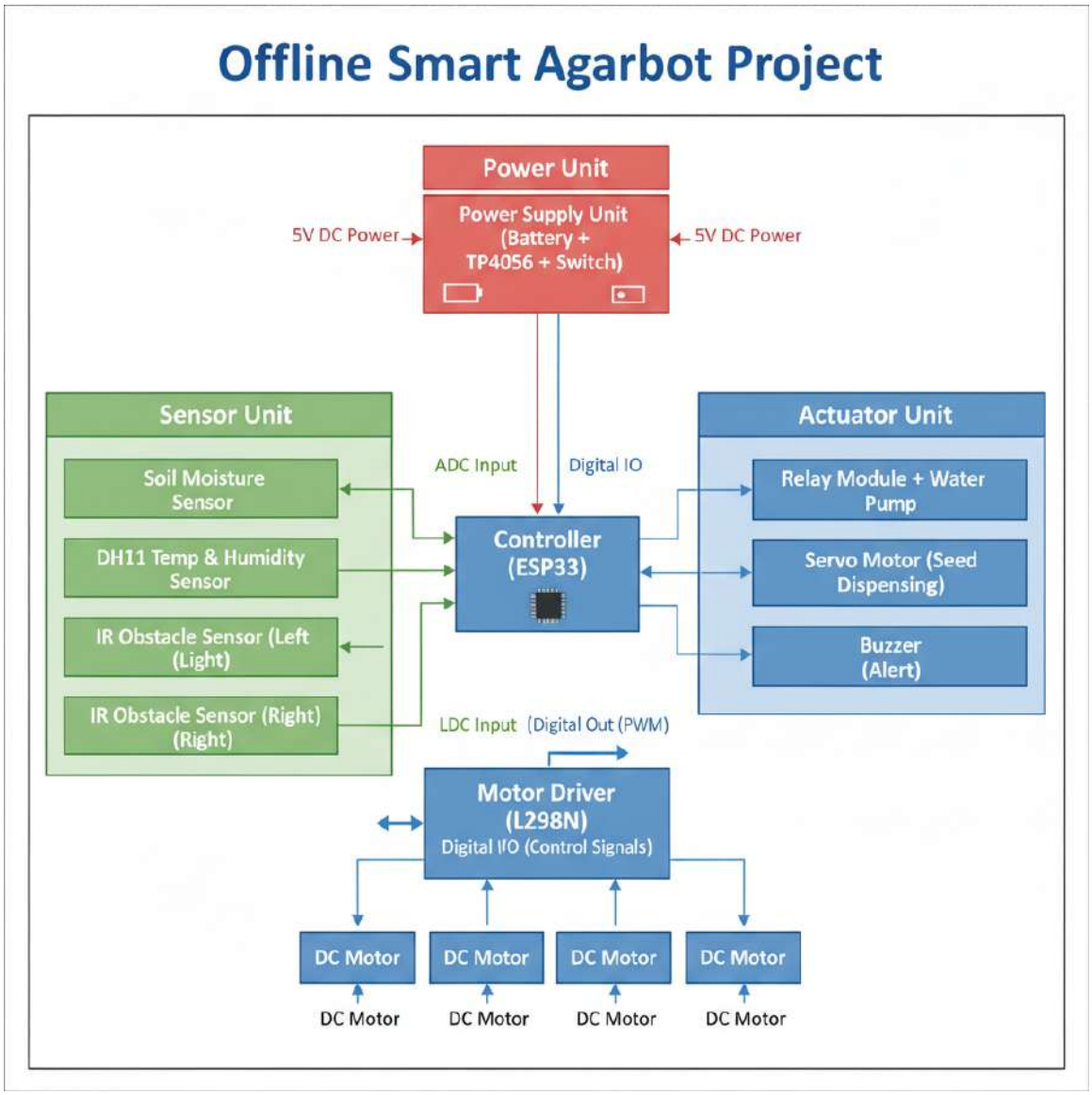
2.3 Objective

To build a unified, mobile, cloud-integrated robotic system that establishes real-time coordination between the field environment and users through the Firebase–MIT App Inventor ecosystem.

3. System Overview and Block Diagram

3.1 System Architecture

The Smart AgroBot architecture integrates sensing modules, a central processing unit (ESP32), actuation systems, and cloud connectivity to form a cohesive IoT ecosystem. The hierarchical design ensures efficient data flow and real-time control capabilities.



4. Hardware Implementation

4.1 Components and Specifications

Component	Function	Specifications
ESP32 Dev Board	Central processing & Wi-Fi	Dual-core MCU, 240 MHz
DHT11 Sensor	Temperature & humidity measurement	0-50C, 20-90% RH
Soil Moisture Sensor	Measures soil water level	Analog output
L298N Motor Driver	Controls DC motor direction	Dual H-bridge, 2A output
Relay Module	Controls pump	5V, opto-isolated
Water Pump	Irrigation system	3-6V DC
Li-ion Battery	Portable power source	3.7V x 2
TP4056 Module	Battery charging	Micro-USB
4WD Chassis	Robotic body	4 BO motors + wheels
Miscellaneous	LEDs, resistors, switches	For indicators and control

5. Software Implementation

5.1 Development Environment

- Platform: Arduino IDE
- Libraries:
  - WiFi.h (Wi-Fi connectivity)
  - FirebaseESP32.h (Firebase communication)
  - DHT.h (sensor data)
  - Servo.h (seed-dispenser control)

5.2 Firebase Database Structure

agrobot/

- ├── temperature
- ├── humidity
- ├── soil_moisture
- ├── pump
- └── movement

5.3 Mobile Application

Developed in MIT App Inventor with two modules:

- Monitoring Dashboard: Displays live sensor readings.
- Control Panel: Sends movement and irrigation commands.

Data is synced bi-directionally with Firebase through real-time database triggers ensuring instant updates.

6. Novelty and Innovation

- Unified integration of sensing, cloud, and control functionalities.
- Real-time Firebase connectivity for global accessibility.
- User-friendly app designed for low-end devices.
- Economical design reducing cost by 60% over traditional systems.
- Adaptable for rural outreach and IoT education.

7. Sustainable Development Goals (SDG) Alignment

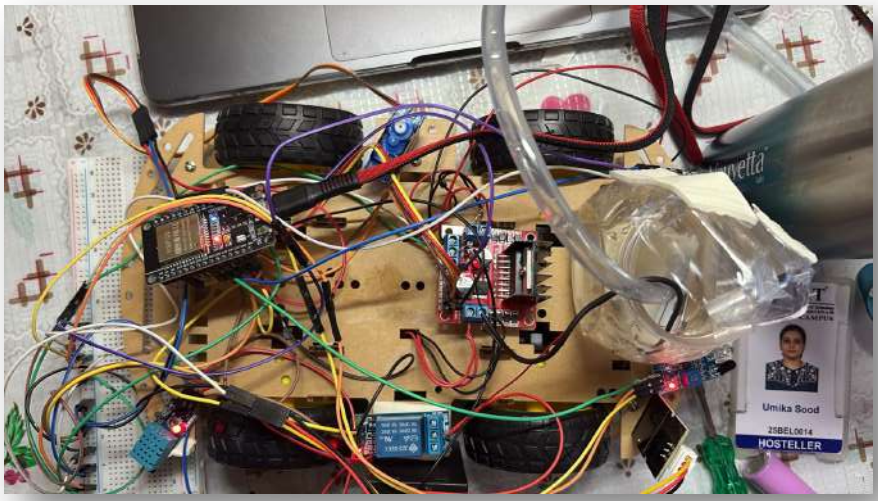
SDG No.	Goal	Contribution
2	Zero Hunger	Promotes sustainable farming and productivity.
6	Clean Water & Sanitation	Enables precision-controlled irrigation.
9	Industry, Innovation & Infrastructure	Advances rural technological infrastructure.
12	Responsible Consumption	Reduces water and energy wastage.
13	Climate Action	Supports climate-resilient agriculture.

8. Experimental Results and Observations

Parameter	Expected Range	Observed Output	Remarks
Soil Moisture	0-1023 (ADC)	340-800	Stable readings
Temperature	25-40C	32C avg	Consistent values
Humidity	40-80% RH	61% avg	Proper sensor calibration
Pump Control	ON/OFF	Responsive	Works in real time
Movement	5-directional	Smooth	Firebase delay <1 s

The AgroBot demonstrated robust data transmission and seamless bidirectional control using Firebase. The overall system responsiveness validated its design goals.

9. Snapshots of Hardware



10. Conclusion and Future Scope

The Smart AgroBot successfully integrates IoT, robotics, and cloud-based data management to address critical agricultural challenges. It provides scalable automation compatible with sustainable development objectives.

Key Achievements:

- Real-time environmental monitoring.
- Wireless robotic control through Firebase.
- Compact, modular, and cost-efficient prototype.

Proposed Enhancements:

- Machine learning for predictive irrigation control.
- Autonomous navigation with GPS and obstacle sensors.
- Solar charging for field endurance.
- NPK sensor integration for nutrient monitoring.
- AI dashboard for multi-robot farm coordination.

11. References

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